

Toolkit Landing Report

RFFPerm / OnlineRFFPerm / PO-learner / FSDS / Multi-layer Feature Graphs
Direct Landing Scenarios for LLM Inference, Alignment, and Data-Quality Consistency

Internal methods briefing (open-source PDFs, shipped code, ready datasets)
Date: 2026-09-15

Abstract

This report inventories an open, implemented toolkit for distribution-shift diagnosis and adaptation, and answers a product question: in LLM **inference/prediction, alignment, hallucination monitoring, synthetic-data quality, and representation consistency**, what can land now, how, and what must not be over-claimed.

Five capability layers: (i) **RFFPerm**—model-agnostic predictive-degradation permutation tests; (ii) **OnlineRFFPerm**—streaming monitoring with online FDR; (iii) **PO-learner / CF-Perm / DRPerm / RRPerm**—treat batches as $T \in \{0, 1\}$ and use causal objective functions as distributional distances plus OOD variable importance; (iv) **FSDS**—feature selection under covariate shift $P(X)$ and concept drift $P(Y | X)$; (v) **multi-layer / multi-scenario / feature-bank graph methods**—ASSOC/HSIC chains and hierarchical subset localization across user–item–session or modality towers.

Non-negotiable stance. These tools **detect, localize, reweight/gate, and schedule compute**. They are **not** fact checkers, and they do **not** uniquely decompose covariate vs. concept to single features without extra structure. KL additivity does not transfer to arbitrary distances; full-space unique attribution is impossible—**subset localization** is the correct production move.

Priority landings (data ready):

1. Streaming hallucination / error-rate **regime** monitors (OnlineRFFPerm + `po_risk0` ranking);
2. Synthetic vs. human preference **quality gates** (RFFPerm-as-Judge);
3. Representation batch geometry (embedding PC1–PC3) before predictive-shift tests;
4. Multimodal towers—*which layer moved* $\rightarrow$ KV refresh / LoRA budget;
5. RecSys feature banks (TencentGR-class): RF-domain for $P(X)$, PO-risk for Y -gap;
6. Gated $\sqrt{\text{PO}}$ reweighting: uniform when quiet; heat on concept hops; anneal.

Contents

1	Problem and boundaries	1
1.1	Four product questions	1
1.2	Claims we refuse	2
1.3	Open-source and local anchors	2
2	Toolkit map: how the five layers lock	2
3	Method cards	3
3.1	RFFPerm	3
3.2	OnlineRFFPerm	3
3.3	PO-learner (DRPerm / RRPerm / CFPerm)	3
3.4	Gated $\sqrt{\text{PO}}$ reweighting (shipped locally)	3
3.5	FSDS	3
3.6	Multi-layer / multi-scenario / feature-bank graphs	3

4	Direct landing scenario matrix (LLM-centric)	4
4.1	A. Inference and prediction monitoring	4
4.2	B. Hallucination and “looks like an answer”	4
4.3	C. Alignment and preference consistency	4
4.4	D. Data quality and representation alignment	5
4.5	E. Multimodal and continuous-inference scheduling	5
4.6	F. Multi-layer business graphs (recs / risk / commerce)	5
5	Ready data and minimum viable experiments	5
6	Scenario × method cheat sheet	6
7	Executable playbooks	6
7.1	Playbook 1: LLM output-quality stream	6
7.2	Playbook 2: Alignment data version gate	6
7.3	Playbook 3: Multimodal continuous inference	6
7.4	Playbook 4: Feature-bank drift workorder	7
8	Engineering conclusions aligned with impossibility	7
9	Risks, fallbacks, MLOps hooks	7
10	Priority roadmap (“can run now”)	7
11	Conclusion	7

1 Problem and boundaries

1.1 Four product questions

1. **Is inference/prediction drifting?** When must a deployed $\hat{f}$ refresh on streaming (X_t, Y_t) ?
2. **Is alignment consistent?** Human prefs, synthetic prefs, judge models, online feedback—same law?
3. **Did hallucination/error enter a new regime?** Not row-wise true/false—did $P(Y | X)$ jump?
4. **Is data quality consistent?** Feature stores, embedding batches, modality towers, synthetic corpora vs. reference.

1.2 Claims we refuse

- No story “feature j causes hallucination.”
- No unique full-space split of $P(X)$ vs. $P(Y | X)$ from raw error alone.
- DRE / PSI / KL are not sufficient statistics for concept drift.
- `po_risk0` is residual strength vs. the previous probe—not a factuality score.

1.3 Open-source and local anchors

Component	Repo / doc
RFPPerm	https://github.com/ruanhq-StatsML/RFPPerm
OnlineRFPPerm	https://github.com/ruanhq-StatsML/OnlineRFPPerm
PO-learner	https://github.com/ruanhq-StatsML/Causal_Objective_Permutation_Test
FSDS	https://github.com/ruanhq-StatsML/FSDS-FeatureSelection_for_DistributionShi
Multi-layer PDF	same repo FSDS_Sep14_MultiLayerFeatureSelectionMetrics.pdf
Gated PO (local)	docs/po_risk/PO_risk_reweight_method.tex
Hallucination proto	scripts/agod/rfperm_hallucination_proto.py (feature branch)

2 Toolkit map: how the five layers lock

Layer	I/O	One-line LLM reading
1. RFPPerm	(X, Y) reference vs. new; reject predictive degradation	“Did this batch of answers/labels get significantly worse?”
2. OnlineRFPPerm	Streaming error p -values + SAF-FRON/ADDIS	“When should we refresh the model/judge online?”
3. PO / R / CF	$T = 0/1$, PO-/R-risk, VIMP	“Mapping shift vs. traffic mix? Which dims carry the gap?”
4. FSDS	Covariate: RF-domain / MMD-LOCO; concept: PermuCATE / DRPerm	“Which feature family moved—fix features or retrain the head?”
5. Multi-layer graph	Families, towers, user-item hierarchy; HSIC/ASSOC chains	“Cross-scenario coupled drift; drill to a subset, then act.”

Default production pipeline.

- Geometry pre-check:** PCA (PC1–PC3) + domain AUC on embeddings—alignment viz, not the final call.
- Regime test:** OnlineRFPPerm (or gated consecutive OOS ratio) decides fire.
- Axis diagnosis:** stop if quiet; if fire, RF-domain ($P(X)$) and PO/R ($P(Y | X)$ proxy) in parallel.
- Feature/layer localization:** FSDS top- k + family mass; multi-layer $\Rightarrow$ leave-one-group-out.
- Action:** quiet $\rightarrow$ uniform; heterogeneous concept $\rightarrow \sqrt{\text{PO}}$ weights; global map flip $\rightarrow$ drop-old; modality drift $\rightarrow$ refresh that tower’s KV / update budget.
- Anneal:** next hop’s e_{prev} is large, ratio falls, return to uniform.

3 Method cards

3.1 RFPPerm

Compares validation/OOB error on a reference batch to predictive error on a new batch via permutation. Density-aware; sensitive to both covariate and concept; does **not** name the axis alone.

LLM use. Prompts/representations as X ; correctness / preference / hallucination labels as Y . Strong at $n \sim 30\text{--}40$; fits per-observation synthetic-data gates (RFPPerm-as-Judge).

3.2 OnlineRFPPerm

Streaming extremity of predictive error vs. the deployed model; online FDR (SAFFRON/ADDIS, async variants) controls false alarms. Modality-agnostic whenever a predictive model and labels (or proxies) exist.

LLM use. Online judge scores, tool-success, refusal rate, hallucination rate as **regime** monitors; optional clever-covariate injection into a light adapter.

3.3 PO-learner (DRPerm / RRPerm / CFPerm)

Old batch $T = 0$, new $T = 1$. Doubly robust / R-learner objectives with permute-then-refit; VIMP reads as **OOD variable importance**. Meta-learners estimate distances, not causal effects. R-risk is more $P(X)$ -insensitive under projection heuristics—a concept-axis proxy, not a uniqueness theorem.

LLM use. Alignment shift (same prompt law, changed preference/correctness map) vs. traffic mix shift. Instance $\text{po_risk}_i = |Y_i - \mu_0(X_i)|$ (or $1 - p$ for class) drives ranking and gated weights.

3.4 Gated $\sqrt{\text{PO}}$ reweighting (shipped locally)

Fire only if consecutive OOS $e_{\text{now}}/e_{\text{prev}} \geq \gamma$ (default 1.5); on fire, $T = 1$ gets $w = \sqrt{\text{po_risk}_0}$ (mean 1); quiet $\Rightarrow w = 1$. Under squared loss, excess risk $\Delta_t^2 = \mathbb{E}_t[(\mu_t^* - \mu_0)^2]$ —the gate opens only when concept OOD is large relative to noise. DRE sees only $p(x)$: near-uniform on pure concept; can hurt on covariate. Two levers: soft **reweight** vs. hard **drop-old**. No always-on IPTW; no ultra-thin PO-tail subsets.

3.5 FSDS

Covariate: RF-domain OOB-VIMP, MMD/HSIC-LOCO, Boruta/ShadowVIMP—strong and often cheaper than FSL-Net/DataFix.

Concept: PermuCATE, DRPerm-LOCO, GRF VIMP, R-risk LOCO, Delta-Shapley, Knockoff/L1—meta-learners shine on nonlinear concept drift.

Reading discipline. Rank VIMP, then post-hoc $\mathbb{E}[X | Q_1]$ vs. $\mathbb{E}[X | Q_4]$; mass + directional means describe “who arrived later,” not causation. Empirically on TencentGR clocks: cross / decay / in-session conversion carry the gap; last-touch attribution families are not the shift source.

3.6 Multi-layer / multi-scenario / feature-bank graphs

Feature stores are hierarchical (user / item / session / merchant; or audio / image / text). Unique single-feature attribution on the full space is impossible; **hierarchical subset localization** is the right move.

ASSOC / HSIC chains. Pairwise HSIC across layered metrics; threshold to edges; connected components are metric chains (e.g., delivery–rating–orders–GMV). Association only—no directed causal claim.

Leave-one-group-out. Block/modality contribution to shift; conditional drill (region then category) avoids confounding levels.

Continuous multimodal inference. FSDS shares π_m decide which tower to re-encode per window, which KV to invalidate, which tower gets test-time update budget; static captions encode-once. Attribution-Guided Online Distillation (AGOD) routes student losses with modality gaps as clever covariates.

4 Direct landing scenario matrix (LLM-centric)

All rows assume: code exists, inputs extractable from current pipes, outputs wire to alerts / gates / reweight / schedulers. ★ = ready in this repo or proto.

4.1 A. Inference and prediction monitoring

Scenario	Primary tools	Landing action
A1 Online task accuracy / tool-success regime jump ★	OnlineRFPerm + gated OOS	fire → alert; online FDR; optional drop-old on judge head
A2 Delayed feedback (like/complaint) concept jump	PO-risk stream + RRPerm	Refresh preference head only if mapping moved; resample if mix moved
A3 Multi-task router misfires	FSDS on task one-hots + reps	High-VIMP task block → retrain that expert only
A4 RAG hit-rate vs. answer correctness	RF-domain (retrieval feats) + PO (correctness)	Retrieval drift ≠ generation drift; fix index vs. generator separately

4.2 B. Hallucination and “looks like an answer”

- **B1 Regime-level hallucination-rate monitor ★.** Synthetic concept cut: quiet then noisy. OnlineRFPerm/gate fires at the cut; instance `po_risk0` ranks audit candidates (proto AUROC ~ 0.63 – 0.64). **Not** fact checking.
- **B2 Hallucination-labeler drift.** Same generations, human/auto Y changes → PO/R; generation mix changes → RF-domain.
- **B3 Refusal / safety-policy hops.** Safety action as Y ; monitor whether the policy head entered a new operating point—do not confuse traffic mix with “model got worse.”

4.3 C. Alignment and preference consistency

- **C1 Human vs. synthetic preference.** RFPerm-as-Judge: small- n test that synthetic prefs are not predictively off the human batch.
- **C2 SFT/RLHF/DPO version gates.** Old vs. candidate preference data: FSDS lists which prompt features carry the gap; fail closed on merge.
- **C3 Judge-model calibration.** Judge score as Y ; OnlineRFPerm vs. a gold set; high PO $\Rightarrow$ sample audit, not full relabel.
- **C4 Instruction-following vs. style covariates.** Style/length/language mix → RF-domain; correctness map → PO. Do not explain accuracy drops with $\text{PSI}(\text{length})$.

4.4 D. Data quality and representation alignment

- **D1 Embedding batch geometry ★.** Point-cloud / text / multimodal embeddings: PC1–PC3 + domain AUC; only then RFPerm/FSDS. Fix the pipe if geometry is broken.
- **D2 Feature-bank health (TencentGR-class) ★.** Combinatorial features → FSDS: RF for cohort portrait, PO for conversion gap; family mass drives feature workorders.

- **D3 Synthetic corpus diversity vs. toxic patterns.** RFPerm on quality labels; FSDS localizes toxic pattern features (length, template n -grams, embedding clusters).
- **D4 Train-serve feature parity.** Same feature names offline vs. online as $T = 0/1$; high RF-domain VIMP = pipeline-bug suspect list.

4.5 E. Multimodal and continuous-inference scheduling

- **E1 Tower attribution (MSR-VTT / ChronoBerg-class) ***. Block FSDS + PO; shares π_v, π_a, π_t set re-encode and update masks.
- **E2 Continuous VLM inference.** Static text encode-once; vision/audio refresh per window; KV invalidation on session-clustered monitors (avoid global pooled cancellation).
- **E3 AGOD distillation routing.** $\alpha_m = \text{softmax}(g_m/\tau)$ with g_m from domain AUC·VIMP + γ ·PO-risk; fall back to RF-domain under poor overlap.

4.6 F. Multi-layer business graphs (recs / risk / commerce)

- **F1 Hierarchical subset localization.** Significant week batch $\rightarrow$ user-region first, then category; emit {subset, split rule, stability} to owners.
- **F2 ASSOC metric chains.** HSIC edges surface cross-layer coupling (fulfillment-rating-orders-GMV); alert on chains, not single KPIs.
- **F3 Fraud mix vs. fraud tactic.** Matches the open PDFs: cohort mix = covariate; tactic = concept; separate tickets.

5 Ready data and minimum viable experiments

Data	Shape		Suggested experiment
data/public/*.npz	Public	time-series	Gated $\sqrt{\text{PO}}$ vs. DRE vs. uniform (board exists)
data/tencent_subset	RecSys	sequences + feature bank	FSDS on time/business clocks; family mass
ChronoBerg / FMoW / WhyShift	Public	FSDS bench-marks	Reproduce covariate/concept feature selection
MSR-VTT packed reps	video	audio text	Modality attribution + AGOD routing
Hallucination synth stream	Proto	script	fire@cut + po_risk0 ranking
OpenML (FSDS)	tabular	8 tabular sets	RF-domain vs. FSL-Net compute compare

Minimum local commands.

```
# Hallucination regime proto (branch cursor/rfperm-hallucination-92f8)
python3 scripts/agod/rfperm_hallucination_proto.py

# Gated PO method notes:
# docs/agod/AGOD_online_rfperm.md
# docs/po_risk/PO_risk_reweight_method.tex

# Tencent feature selection (time clock)
python3 scripts/tencent_gr/run_fsds_time.py
```

6 Scenario × method cheat sheet

Scenario cluster	RFPerm	Online	PO/R	FSDS/graph
Inference regime monitor	•	•	•	
Hallucination candidate rank		•	•	
Alignment / pref gates	•	•	•	•
Representation geometry	•			•
Feature-bank quality			•	•
Multimodal scheduling		•	•	•
Multi-layer business drill			•	•

•=primary; •=support.

7 Executable playbooks

7.1 Playbook 1: LLM output-quality stream

1. Define $Y \in \{0, 1\}$: correct / non-hallucinated / preference win (labels or weak supervision).
2. X = prompt embedding + light metadata (length, tool calls).
3. Fit shallow RF probe μ_0 on reference window; stream $e_{\text{now}}/e_{\text{prev}}$.
4. On fire: alert + Top- k audit by `po_risk0`; optional $\sqrt{\text{PO}}$ refit of a light head.
5. Attach online FDR so diurnal traffic jitter does not page on-call.

7.2 Playbook 2: Alignment data version gate

1. Old preference set $T = 0$; candidate merge set $T = 1$.
2. RF-domain: very high AUC with top feature “source=synthetic” $\Rightarrow$ portrait contamination.
3. DRPerm/PermuCATE: significant concept axis $\Rightarrow$ preference map moved; human review, no silent merge.
4. Merge only if both axes quiet (or only explained covariates with owner sign-off).

7.3 Playbook 3: Multimodal continuous inference

1. Per window extract X_v, X_a, X_t ; block RF-domain + PO vs. reference.
2. Obtain π_m ; if $\pi_t \approx 0$, caption encode-once.
3. High $\pi_v \Rightarrow$ refresh vision KV / raise vision update steps.
4. Spurious text VIMP rise is a **leakage diagnostic**, not a reason to spend budget on the language tower.

7.4 Playbook 4: Feature-bank drift workorder

1. Fix a business clock (time quantiles / experiment buckets).
2. Emit RF-mass and PO-mass by family with Q1→Q4 means.
3. Families with negative LOGO Δ are not “root causes.”
4. Ticket only high-mass families with readable direction (decay clicks, in-session conversion, crosses).

8 Engineering conclusions aligned with impossibility

1. **Unique error decomposition is impossible:** model change vs. data change without structure.
2. **Unique axis decomposition is impossible:** except special divergences (e.g., KL), raw error does not uniquely split $P(X)$ vs. $P(Y | X)$.
3. **Unique single-feature decomposition is impossible:** ship VIMP + subset localization; never “feature j is 23% of the drift” as a KPI.
4. **What remains feasible:** jointly sensitive tests (RFPerm); axis proxies (R/PO vs. domain); subset/layer localization; gated adaptation.

9 Risks, fallbacks, MLOps hooks

- **Poor overlap:** unstable meta-learners—fall back to RF-domain / refuse fire / widen reference.
- **Empty-room denominators:** classification $e_{\text{prev}} \approx 0$ false-fires—set e_{floor} .
- **Always-on IPTW:** loses to uniform on quiet clocks—default uniform; heat only on gate.
- **Parallelism:** permutation tests parallelize; literature cites $\sim 60\text{--}70\%$ wall-clock cuts.
- **Hooks:** feature-quality boards, merge gates, online alerts, multimodal schedulers, synthetic-data acceptance.

10 Priority roadmap (“can run now”)

1. **P0 (this week):** keep public-clock gated-PO board; wire hallucination proto to real judge labels; freeze TencentGR FSDS workorder template.
2. **P1:** alignment data version gate (human vs. synthetic prefs) in CI.
3. **P1:** MSR-VTT / multimodal $\pi_m \rightarrow$ inference scheduler config.
4. **P2:** ASSOC chains onto business KPI graphs.
5. **P2:** Online FDR onto the production alert bus.

11 Conclusion

RFPerm / OnlineRFPerm / PO-learner / FSDS / multi-layer graphs already form a closed loop: **test** $\rightarrow$ **localize** $\rightarrow$ **gated adapt** $\rightarrow$ **schedule**. In LLM settings the valuable artifact is not “another hallucination score,” but:

With data and labels ready: regime detection under statistical gating, interpretable feature/modality localization, spend reweighting and compute only on true fires, and anneal on the next hop.

FSDS is the default diagnostic layer for **feature-bank and multi-tower consistency**; PO-learner for **alignment maps** and **concept hops**; OnlineRFPerm for **online regimes**; multi-layer ASSOC for **cross-scenario drills**. Keep “detection $\neq$ causation, localization $\neq$ unique decomposition,” and the toolkit ships to production instead of staying in paper figures.

Chinese edition: LLM_FSDS_Landing_Report_CN.tex.

Build

```
cd docs/reports
xelatex LLM_FSDS_Landing_Report_CN.tex
pdflatex LLM_FSDS_Landing_Report_EN.tex
```